

HCSM-11

The 5 + 9 Split of the 14-Mode Multiplet

Substrate Eigenmodes, Mesh Resonances, and the $\delta = N/4 - \text{base}$ Classification The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model derives the 14-mode multiplet at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus as one Standard Model generation (HCSM-04). This paper establishes the 5 + 9 split of this multiplet into 5 *Substrate Eigenmodes* and 9 *Mesh Resonances*. The classification is defined by the quantity $\delta = N/4 - \text{base}$, where $N/4$ is the mode particle calibration label (framework content, the subject of HCSM-23) and base is a framework base formula expressed in the framework constants d and H . The paper proves that exactly 5 of the 14 modes have $\delta = 0$ (the Substrate Eigenmodes) and exactly 9 have $\delta \neq 0$ (the Mesh Resonances). The 5 Substrate Eigenmodes are the dark photon, the τ lepton, the W boson, the Z boson, and the Higgs boson: the 4 bosons of the multiplet together with τ . The 9 Mesh Resonances are the remaining 9 fermions. The split is proved conditional on the framework's base formula assignment and the $N/4$ calibration table; both are stated as inputs. All claims are HCSM-native and verified at machine precision.

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1 Introduction

1.1 The 5 + 9 split

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation (HCSM-04 [3]). Each mode is assigned to a Standard Model particle, and each particle carries an $N/4$ label that orders it by mass (HCSM-23 [8]).

This paper establishes a structural classification of the 14 modes: the 5 + 9 *split* into 5 *Substrate Eigenmodes* and 9 *Mesh Resonances*. The classification is defined by the quantity

$$\delta = N/4 - \text{base}, \quad (1)$$

where base is a framework base formula expressed in the constants d and H . Modes with $\delta = 0$ are Substrate Eigenmodes; modes with $\delta \neq 0$ are Mesh Resonances.

The split is the subject of this paper. It is analogous in structure to the sector-coarse correspondence of HCSM-06 [?] and the hypercharge quantization of HCSM-07 [?]: a structural classification of the multiplet that is conditional on the framework's calibration input.

1.2 The one line summary

The 14 modes at $\lambda = 4$ decompose into 5 Substrate Eigenmodes (those whose $N/4$ labels equal a framework base formula exactly) and 9 Mesh Resonances (those whose $N/4$ labels are offset from the base formula). The 5 eigenmodes are the 4 bosons of the multiplet plus τ ; the 9 resonances are the remaining 9 fermions.

1.3 Main results

- (1) **The δ function** (Definition 3.1). The quantity $\delta_m = N/4_m - \text{base}_m$ is defined for each of the 14 modes; $N/4_m$ is the calibration label and base_m is the framework base formula.
- (2) **The δ table** (Lemma 3.2). The δ values for the 14 modes are explicitly

$$\delta = (33, +6, +3, +3, -3, -3, 0, +0.5, 0, +0.5, 0, 0, 0, +0.5) \quad (2)$$

in standard mode order.

- (3) **The 5+9 Split Theorem** (Theorem 4.2). Exactly 5 modes have $\delta = 0$; exactly 9 have $\delta \neq 0$.
- (4) **Eigenmode identification** (Corollary 4.3). The 5 Substrate Eigenmodes are dark photon, τ , W , Z , H .
- (5) **Resonance identification** (Corollary 4.4). The 9 Mesh Resonances are ν , e , u , d , s , μ , c , b , t .
- (6) **Boson relation** (Corollary 5.1). The 5 Substrate Eigenmodes are exactly the 4 bosons of the multiplet (dark, W , Z , H) together with τ . Equivalently, the eigenmode set is $B \cup \{\tau\}$ where $B = \{\text{dark}, W, Z, H\}$ is the boson sector.
- (7) **Relation to D_4 orbits and F/B split** (Section 5). The 5 + 9 split does not coincide with the D_4 orbit decomposition of HCSM-05 nor with the fermion/boson split of HCSM-13; it is a distinct structural classification.

1.4 Cross references to the HCSM paper stack

The paper depends on the following companions in the HCSMstack:

- HCSM-00 [1]: foundations, notation, and mapping.
- HCSM-01 [2]: the 8×8 torus substrate; the framework constants.
- HCSM-04 [3]: the 14-mode multiplet.
- HCSM-05 [4]: orbit decomposition.
- HCSM-13 [5]: the processor T ; the F/B split.
- HCSM-23 [8]: the $N/4$ labels (calibration content).

1.5 Organization

Section 2 states the 14-mode multiplet, the $N/4$ labels (input), and the framework base formulas (input). Section 3 defines the δ function and computes its values. Section 4 proves the 5 + 9 Split Theorem. Section 5 characterizes the eigenmodes and the resonances and relates them to D_4 orbits and the F/B split. Section 6 summarizes inputs and derived results. Section 7 gives reproducibility notes. Section 8 concludes.

2 Setup

2.1 The 14-mode multiplet

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites (HCSM-01 [2]). The 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian are (HCSM-04 [3])

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (3)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (4)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (5)$$

The standard mode order, matching the zero-state vector of HCSM-22 [7] and the mass-ordering labels of HCSM-23 [8], is

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (6)$$

with the correspondence

$$\begin{array}{llll} \nu : (0, 4), & e : (4, 0), & u : (1, 3), & d : (1, 5), \\ s : (3, 1), & \mu : (3, 7), & \text{dark} : (5, 7), & c : (7, 3), \\ \tau : (7, 5), & b : (5, 1), & W : (2, 2), & Z : (2, 6), \\ H : (6, 2), & t : (6, 6). & & \end{array}$$

The identification of the 14 modes with the 14 Standard Model particles is an empirical input of the framework (HCSM-04, Section 2); it is not derived.

2.2 Framework constants

The framework constants used in this paper are

$$d = 4, \quad H = 2^d = 16, \quad N = d \cdot 2^d = 64, \quad V = (d - 1) \cdot 2^d = 48. \quad (7)$$

These are derived in HCSM-01 [2].

2.3 The $N/4$ labels

Each mode carries an $N/4$ label (HCSM-23 [8]). In the standard mode order (6), the labels are

$$(N/4)_i = (60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5). \quad (8)$$

The $N/4$ labels are the framework's calibration table for the particle spectrum; their derivation from the substrate geometry is the subject of HCSM-23 [8]. In the present paper, they are treated as inputs.

2.4 The framework base formulas

Each particle is assigned a *framework base formula*, an expression in the framework constants d and H . The assignment is part of the framework's structural content; it is an input to this paper, not a derivation.

Definition 2.1 (Framework base formula). The framework base formula for each Standard Model particle is

Particle	Base formula	Base value (at $d = 4$, $H = 16$)
ν	$H + 2d + 3$	27
e	$H + d - 2$	18
u	$H + 2$	18
d	$H + 1$	17
s	$H + 1$	17
μ	$H + 1$	17
dark	$H - d$	12
c	$H + 1 - 2d$	9
τ	$H + 1 - 2d$	9
b	$H + 3 - 3d$	7
W	$d/2$	2
Z	$d/2$	2
H_{Higgs}	$H + 1 - d^2$	1
t	$H - d^2$	0

Remark 2.2 (The W/Z correction). The base formula for W and Z is $d/2 = 2$. An earlier framework version used $2(H + 2) - d^2 = 20$, which did not match the observed $N/4$ value of 2. The present paper uses the corrected formula $d/2$, consistent with the framework’s corrected calibration.

Remark 2.3 (Framework formulas). Each base formula is a polynomial in d with coefficients in $\{1, 2, 3\}$ and additive constant in $\{-2, -1, 0, 1, 2, 3\}$, or involves $H = 2^d$ linearly (with an occasional d^2 term for the heavy bosons). The formulas are HCSM-native expressions in the framework constants d and H . Their assignment to specific particles is part of the framework’s structure, treated here as input.

3 The δ function

Definition 3.1 (δ). For each mode m of the 14-mode multiplet, the *deviation* δ_m is

$$\delta_m = N/4_m - \text{base}_m, \quad (9)$$

where $N/4_m$ is the calibration label (8) and base_m is the framework base value (Definition 2.1).

Lemma 3.2 (δ table). *In the standard mode order (6), the δ values are*

<i>Particle</i>	(n_1, n_2)	$N/4$	<i>base</i>	δ
ν	(0, 4)	60	27	+33
e	(4, 0)	24	18	+6
u	(1, 3)	21	18	+3
d	(1, 5)	20	17	+3
s	(3, 1)	14	17	-3
μ	(3, 7)	14	17	-3
dark	(5, 7)	12	12	0
c	(7, 3)	9.5	9	+0.5
τ	(7, 5)	9	9	0
b	(5, 1)	7.5	7	+0.5
W	(2, 2)	2	2	0
Z	(2, 6)	2	2	0
H_{Higgs}	(6, 2)	1	1	0
t	(6, 6)	0.5	0	+0.5

Equivalently, in vector form

$$\delta = (+33, +6, +3, +3, -3, -3, 0, +0.5, 0, +0.5, 0, 0, 0, +0.5). \quad (10)$$

Proof. Direct computation: $\delta_m = N/4_m - \text{base}_m$ for each of the 14 modes, using the $N/4$ labels (8) and the base values from Definition 2.1. $\square$

Remark 3.3 (δ values are integers or half-integers). Every δ value in the table is an integer or a half integer. The nonzero values are $\{+33, +6, +3, -3, +0.5\}$ up to signs.

4 The 5 + 9 Split Theorem

Definition 4.1 (Substrate Eigenmode and Mesh Resonance). A mode m of the 14-mode multiplet is a *Substrate Eigenmode* if $\delta_m = 0$; it is a *Mesh Resonance* if $\delta_m \neq 0$.

Theorem 4.2 (5 + 9 Split). *Of the 14 modes at $\lambda = 4$, exactly 5 are Substrate Eigenmodes and exactly 9 are Mesh Resonances.*

Proof. By Definition 4.1, the Substrate Eigenmodes are the modes with $\delta_m = 0$. From Lemma 3.2, the modes with $\delta = 0$ are

$$\{\text{dark}, \tau, W, Z, H_{\text{Higgs}}\}, \quad (11)$$

which has cardinality 5. The remaining $14 - 5 = 9$ modes have $\delta \neq 0$; they are the Mesh Resonances. $\square$

Corollary 4.3 (Eigenmode identification). *The 5 Substrate Eigenmodes are the dark photon, the τ lepton, the W boson, the Z boson, and the Higgs boson.*

Proof. Immediate from the proof of Theorem 4.2. $\square$

Corollary 4.4 (Resonance identification). *The 9 Mesh Resonances are the neutrino ν , the electron e , the up quark u , the down quark d , the strange quark s , the muon μ , the charm quark c , the bottom quark b , and the top quark t .*

Proof. Immediate from the proof of Theorem 4.2 and the resonance set $M \setminus \{\text{dark}, \tau, W, Z, H_{\text{Higgs}}\}$. $\square$

5 Structure of the split

5.1 The Substrate Eigenmodes and the boson sector

Corollary 5.1 (Boson relation). *Let $B = \{\text{dark}, W, Z, H_{\text{Higgs}}\}$ denote the boson sector of the multiplet (the kernel of the processor T of HCSM-13 [5]). Let $F = \{\nu, e, u, d, s, \mu, c, \tau, b, t\}$ denote the fermion sector (the image of T). Then*

$$\{\text{Substrate Eigenmodes}\} = B \cup \{\tau\}, \quad (12)$$

$$\{\text{Mesh Resonances}\} = F \setminus \{\tau\}. \quad (13)$$

Equivalently, the Substrate Eigenmodes consist of the 4 bosons of the multiplet together with τ ; the Mesh Resonances are the 9 fermions other than τ .

Proof. By Corollary 4.3, the Substrate Eigenmodes are $\{\text{dark}, \tau, W, Z, H_{\text{Higgs}}\}$. Of these, $\text{dark}, W, Z, H_{\text{Higgs}}$ are the boson sector B (they are in the kernel of the processor T of HCSM-13 [5]); τ is a fermion. Hence the eigenmodes are $B \cup \{\tau\}$.

By Corollary 4.4, the Mesh Resonances are $\{\nu, e, u, d, s, \mu, c, b, t\}$, which is $F \setminus \{\tau\}$. $\square$

Remark 5.2 (τ as the exceptional fermion). The 5 + 9 split singles out τ as the unique fermion in the Substrate Eigenmode set. Among the 10 fermions, 9 are Mesh Resonances and 1 (τ) is a Substrate Eigenmode. The framework does not at this level explain why τ is distinguished; the observation is structural.

5.2 Relation to D_4 orbits

The 5 + 9 split does not coincide with the D_4 orbit decomposition of HCSM-05 [4], in which the 14 modes split into orbits of sizes 2, 8, 4.

Orbit	Modes	δ values
$\mathcal{O}_0$	$\{\nu, e\}$	+33, +6
$\mathcal{O}_1$	$\{u, d, s, \mu, \text{dark}, c, \tau, b\}$	+3, +3, -3, -3, 0, +0.5, 0, +0.5
$\mathcal{O}_2$	$\{W, Z, H, t\}$	0, 0, 0, +0.5

The Substrate Eigenmodes are distributed across orbits: $\mathcal{O}_0$ contains 0 eigenmodes, $\mathcal{O}_1$ contains 2 (dark, τ), and $\mathcal{O}_2$ contains 3 (W, Z, H). The Mesh Resonances are distributed across orbits: $\mathcal{O}_0$ contains 2, $\mathcal{O}_1$ contains 6, and $\mathcal{O}_2$ contains 1.

The 5 + 9 split is therefore a distinct structural classification, not a coarse-graining of the D_4 orbit decomposition.

5.3 Relation to the winding class partition

The winding class partition (HCSM-13 [5], HCSM-14 [6]) has three classes:

$$A = \{\nu, e, t\}, \quad B = \{u, d, s, \mu, c, \tau, b\}, \quad C = \{\text{dark}, W, Z, H\}. \quad (14)$$

The Substrate Eigenmodes are distributed across the classes: A contains 0 eigenmodes, B contains 1 (τ), C contains 4 (dark, W, Z, H). The Mesh Resonances are distributed: A contains 3, B contains 6, C contains 0.

The 5 + 9 split also does not coincide with the winding class partition.

5.4 The W/Z correction

The framework base formulas for the W and Z bosons are $d/2 = 2$ (Remark 2.2). This is consistent with the observed $N/4 = 2$ for both, giving $\delta = 0$. An earlier framework version used the formula $2(H + 2) - d^2 = 20$ for the same particles, giving $\delta = 2 - 20 = -18 \neq 0$. The correction to $d/2$ is the structural condition that both W and Z are Substrate Eigenmodes with $\delta = 0$; the corrected formula $d/2$ is consistent with this, and its assignment to W and Z is part of the framework's corrected base formula table.

6 Inputs and derived

6.1 Derived

The following results are proved in this paper:

1. The definition of the δ function (Definition 3.1).
2. The δ values for the 14 modes (Lemma 3.2).
3. The 5 + 9 Split Theorem (Theorem 4.2).
4. The identification of the 5 Substrate Eigenmodes (Corollary 4.3).
5. The identification of the 9 Mesh Resonances (Corollary 4.4).
6. The boson relation $\{\text{eigenmodes}\} = B \cup \{\tau\}$ (Corollary 5.1).

6.2 Inputs

1. **The 14 mode multiplet** (HCSM-04 [3]).
2. **The mode particle identification**: the assignment of each of the 14 modes to a Standard Model particle. This is an empirical input of the framework (HCSM-04).
3. **The $N/4$ labels** (8): the framework calibration content (HCSM-23 [8]). Treated as input here.
4. **The framework base formulas** (Definition 2.1): the polynomials in d and H assigned to each particle. Treated as input here.
5. **The framework constants** $d = 4$, $H = 16$, $N = 64$ (HCSM-01 [2]).

6.3 What is not claimed

We do not claim to derive the framework base formulas from first principles, nor to derive the $N/4$ labels from the substrate geometry. These are inputs from the framework's structure. The paper's content is: given the base formulas and the $N/4$ labels, exactly 5 of the 14 modes have $\delta = 0$ (the Substrate Eigenmodes) and 9 have $\delta \neq 0$ (the Mesh Resonances).

The framework's claim is: the 5 + 9 split is a structural feature of the 14-mode multiplet with respect to the framework's calibration input.

7 Reproducibility

All numerical claims are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSMcompanion repository at `12-VERIFICATION/HCSM-11_verification.py`. It constructs the $N/4$ table and the base formula table, computes $\delta_m = N/4_m - \text{base}_m$, and verifies:

1. The δ table (Lemma 3.2).
2. The 5 + 9 Split Theorem (Theorem 4.2).
3. The eigenmode identification (Corollary 4.3).
4. The resonance identification (Corollary 4.4).
5. The boson relation (Corollary 5.1).

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSMpaper stack.

8 Conclusion

This paper has established the 5 + 9 split of the 14-mode multiplet of the Hodge Complex Standard Model.

The split is defined by the quantity $\delta = N/4 - \text{base}$, where $N/4$ is the calibration label (framework content) and base is the framework base formula (polynomial in the framework constants d and H). Exactly 5 of the 14 modes have $\delta = 0$ (the Substrate Eigenmodes); exactly 9 have $\delta \neq 0$ (the Mesh Resonances).

The 5 Substrate Eigenmodes are the dark photon, the τ lepton, the W boson, the Z boson, and the Higgs boson: the 4 bosons of the multiplet together with τ . The 9 Mesh Resonances are the remaining 9 fermions.

The split does not coincide with the D_4 orbit decomposition of HCSM-05 nor with the fermion/boson split of HCSM-13. It is a distinct structural classification of the 14-mode multiplet.

The framework’s interpretation is: the 5 Substrate Eigenmodes are the modes whose $N/4$ labels are exactly the framework base formulas; the 9 Mesh Resonances are the modes whose $N/4$ labels are offset from the base formulas by a deviation $\delta \neq 0$. The 5 modes are the “pure substrate” content of the multiplet; the 9 modes are the “mesh-processed” content.

References

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- [3] S. Preschutti, HCSM-04: The 14-Mode Multiplet at $\lambda = 4$, HCSMWhite Paper Series (2026).
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- [8] S. Preschutti, HCSM-23: The δ Function and $N/4$ Labels, HCSMWhite Paper Series (2026).

A Web Crawler Metadata

A.1 Dublin Core

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DC.format: application/pdf; text/x-latex
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  Substrate Eigenmodes; Mesh Resonances; framework base formulas;
  N/4 labels; delta classification; 14-mode multiplet
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  equal to zero (Substrate Eigenmodes) and exactly 9 have  $\delta$  not
  equal to zero (Mesh Resonances). The 5 Substrate Eigenmodes are the
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A.2 Schema.org JSON-LD

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A.3 BibTeX for Automated Indexing

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    Eigenmodes) and 9 have  $\delta \neq 0$  (Mesh Resonances). The
    5 Substrate Eigenmodes are the 4 bosons of the multiplet plus  $\tau$ ;
    the 9 Mesh Resonances are the remaining 9 fermions.}
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